

# bim2sim: A Python Framework for Automated Generation of Multi-Domain Building Simulation Models from IFC Data

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## Software

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## Summary

Building Information Modeling (BIM) offers comprehensive data about buildings, but transforming this information into domain-specific simulation models remains challenging. **bim2sim** addresses this gap by providing a Python framework that transforms Industry Foundation Classes (IFC) models into simulation models for multiple domains. This open-source framework implements a two-stage approach with a uniform meta-structure for IFC data extraction and domain-specific plugins for simulation model generation. The framework focuses on Building Energy Performance Simulation (BEPS) and Heating, Ventilation, and Air Conditioning (HVAC) simulations, with support for Computational Fluid Dynamics (CFD) and Life Cycle Assessment (LCA).

## Statement of Need

Energy-efficient building design and operation heavily rely on BEPS, but creating these simulation models manually is time-consuming, error-prone, and requires specialized expertise. While BIM provides rich building data, direct use for simulation faces several challenges:

- Geometric Discrepancies:** Architectural BIM models employ geometric representations that differ from simulation requirements
- Semantic Gaps:** Critical simulation parameters are often missing in BIM models
- Topological Gaps:** HVAC simulation requires accurate component connections often missing in BIM files
- Data Format Incompatibilities:** Simulation domains demand specific input formats different from BIM exports

Creating simulation models manually can take days to weeks for complex buildings, creating a major bottleneck in the design process.

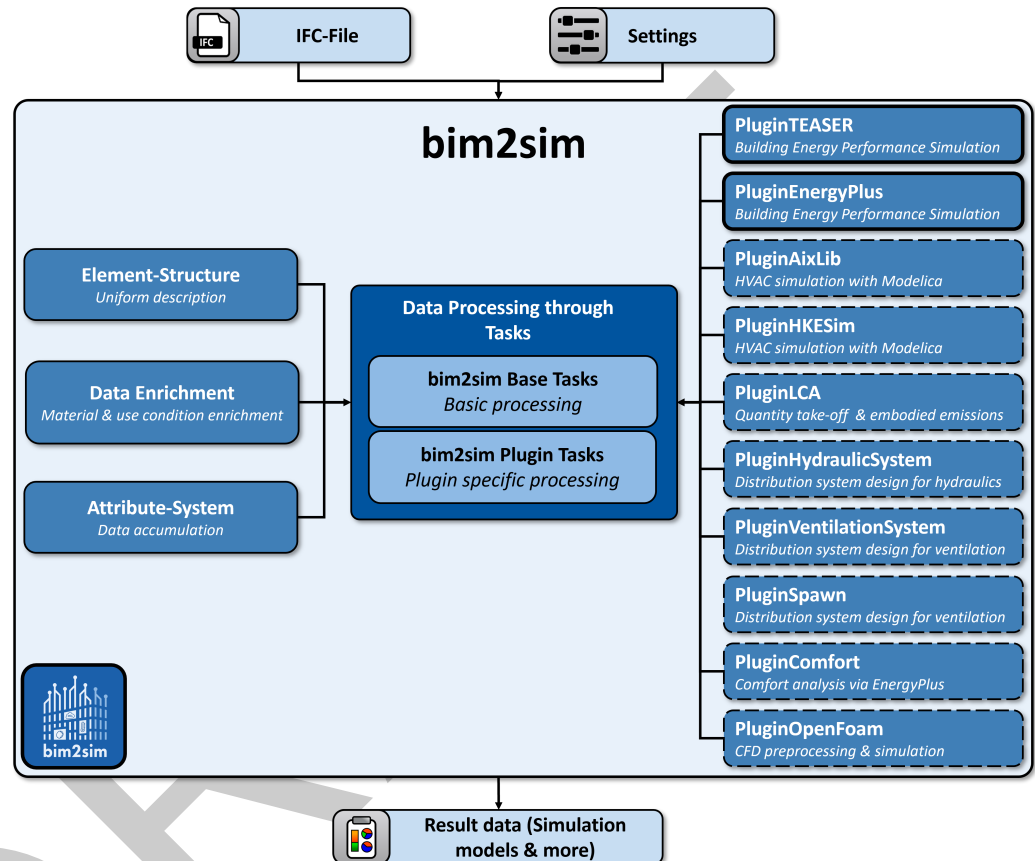
**bim2sim** addresses these needs by:

- Automating transformation from IFC to simulation models, reducing creation time from days to under an hour
- Providing a flexible Python framework extensible to new simulation tools
- Supporting multiple domains through specialized plugins
- Handling imperfect IFC data through repair algorithms
- Maintaining OpenBIM compatibility with minimal dependencies

The framework enables efficient incorporation of building performance simulation into workflows, supporting better-informed decisions.

## Architecture and Implementation

bim2sim is implemented in Python with a two-stage architecture illustrated in Figure 1.



**Figure 1:** bim2sim framework structure and plugins. Dashed plugins are still under heavy active development.

1. **Base Framework:** Transforms IFC data into a uniform meta-structure
2. **Domain-Specific Plugins:** Convert the meta-structure into simulation-ready models

### Core Components of Base Framework

- **Elements:** Domain-specific meta-structure for building component classes
- **Tasks:** Modular processing steps sequenced into workflows
- **Playground:** Task execution environment managing transformations
- **Simulation Settings:** Configuration system for customizing parameters
- **Plugins:** Domain-specific extensions for different simulation targets

### Key Features

- IFC parser utilizing IfcOpenShell (Krijnen, 2018)
- Decision management for handling ambiguities
- Enrichment processes for adding missing information

- 55     ▪ Algorithms to correct Space Boundary (SB) information
- 56     ▪ HVAC system simplification for Modelica simulations
- 57     ▪ Exporters for different simulation platforms

## 58     Available Plugins

59     **bim2sim** currently includes the following plugins:

- 60         1. **TEASER**: Modelica based BEPS simulation using TEASER ([Remmen et al., 2018](#))
- 61         2. **EnergyPlus**: BEPS Simulation using EnergyPlus ([Crawley et al., 2001](#))
- 62         3. **AixLib**: Modelica-based HVAC simulation with AixLib library ([Maier et al., 2023](#))
- 63         4. **HKESim**: Modelica-based HVAC simulation (Modelica library itself is not public available)
- 64         5. **LCA**: Life cycle assessment via quantity takeoff
- 65         6. **Comfort**: Thermal comfort analysis using EnergyPlus

66     The following plugins are currently in development and exist only in feature branches, pending  
67     public release:

- 68         7. **Ventilation & Hydraulic System**: Automatic design of ventilation and hydraulic  
69             distribution systems
- 70         8. **PluginSpawn**: Dynamic coupled simulations of building and HVAC via SpawnOfEnergyPlus  
71             ([Wetter et al., 2024](#))
- 72         9. **OpenFOAM**: CFD simulation (under development, open-source available in fall 2025)

## 73     Existing Publications on Methodology

- 74         ▪ The methodology for **TEASER** and **EnergyPlus** plugins will be published in Jansen et al.  
75             ([Jansen et al., 2025](#))
- 76         ▪ Implementation of **AixLib** and **HKESim** plugins are documented in Jansen et al. ([Jansen  
77             et al., 2023](#))
- 78         ▪ Algorithms for handling SBs are presented in Richter et al. ([V. Richter et al., 2021](#))
- 79         ▪ The **Comfort** plugin framework is presented in Richter et al. ([V. Richter et al., 2023](#))  
80             and evaluated in Richter et al. ([V. E. Richter et al., 2023](#))
- 81         ▪ The methodology of **OpenFOAM** plugin has been presented by Richter et al. ([V. E.  
82             Richter et al., 2024](#)) and extended by Hochberger et al. ([Hochberger & Richter, 2024](#))

## 83     Comparison with Similar Tools

84     **bim2sim** addresses the challenge of leveraging BIM data for building energy simulations, a  
85     field with several existing approaches. In our paper ([Jansen et al., 2025](#)) (currently under  
86     review), we conducted a comprehensive analysis of these BIM-to-simulation tools. [Table 1](#)  
87     presents an abbreviated comparative overview of these tools. The complete analysis in the  
88     to-be-published paper considers additional dimensions such as IFC version support, space and  
89     surface boundary handling, and data enrichment methods. In the abbreviated version shown  
90     here, we focus on the most important aspects: simulation domains (BEPS, HVAC), modular  
91     architecture, open-source availability, and implementation technologies.

**Table 1:** IFC-based approaches from related research (chronologically ordered) as analyzed in ([Jansen et al., 2025](#)). P: Partially, Y: Yes, -: No/not applicable, EP: EnergyPlus, Mod: Modelica, OS: Open-source, Mod.: Modularity, Impl.: Implementation language/framework.

| Reference                                  | Name     | BEPS | HVAC | Mod. | OS | Impl. |
|--------------------------------------------|----------|------|------|------|----|-------|
| ( <a href="#">Bazjanac, 2008</a> )         |          | EP   | -    | Y    | -  | -     |
| ( <a href="#">O'Donnell et al., 2011</a> ) | SimModel | EP   | -    | Y    | -  | XML   |

| Reference                                          | Name              | BEPS         | HVAC     | Mod. | OS | Impl.         |
|----------------------------------------------------|-------------------|--------------|----------|------|----|---------------|
| (Emira El Asmi et al., 2015)                       |                   | COMETH       | COMETH   | -    | -  | -             |
| (G. I. Giannakis et al., 2015)                     |                   | EP, TRNSYS   | -        | -    | -  | Matlab        |
| (Cao, 2018)                                        | SimModel+         | Mod          | Mod      | Y    | -  | Py, C++       |
| (Andriamamonjy et al., 2018)                       | Ifc2ModelicaMod   | Mod          | Mod      | P    | -  | Py, IFC       |
| (Nytsch-Geusen et al., 2017, 2019)                 | CoTeTo            | Mod          | -        | -    | Y  | Py, JModelica |
| (G. Giannakis et al., 2019)                        |                   | SimModel, EP | -        | -    | -  | -             |
| (Ramaji et al., 2020)                              | OsmSerializeOS/EP | OS/EP        | -        | -    | Y  | Java          |
| (W. Chen et al., 2021)                             |                   | EP, eQuest   | Y        | -    | -  | Java          |
| (SIM-VICUS - NEXT LEVEL BUILDING SIMULATION, 2023) | SIM-VICUS         | Nandrad, EP  | district | Y    | Y  | C++           |
| (Z. Chen et al., 2023)                             | AutoBPS-BIM       | EP           | EP       | -    | -  | -             |
| (Mediavilla et al., 2023)                          |                   | EP (not run) | -        | -    | -  | -             |
| bim2sim                                            | bim2sim           | EP, Mod      | Mod      | Y    | Y  | Py, IFC       |

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